

PURPOSE

Can we predict whether a song will reach the Top 10 after just 2-3 weeks on Spotify charts? We build a two-stage pipeline:

- 1. Fit **AR/MA** models to early streaming data to extract momentum features
- 2. Combine trajectory features with audio attributes in a **Bayesian logistic regression** to output P(Top 10)

BACKGROUND

Prior work either uses a song's full streaming history (only available in hindsight) or static audio features alone (ignores early momentum).

We chose two Kaggle datasets: [Spotify Charts](#) (26.2M daily entries, 2017-2021) and [Spotify Tracks](#) (114K songs with audio features). The cleaned dataset contains 3,255 songs with 21-day stream windows.

METHODS

AR/MA Time Series using **Python**

- Extract first 21 days of daily streams per song
- Apply first-order differencing to remove trend
- ACF/PACF analysis across 200 songs confirms ARMA(1,1)
- Fit AR(1) **manually** via OLS
- Fit MA(1) via statsmodels

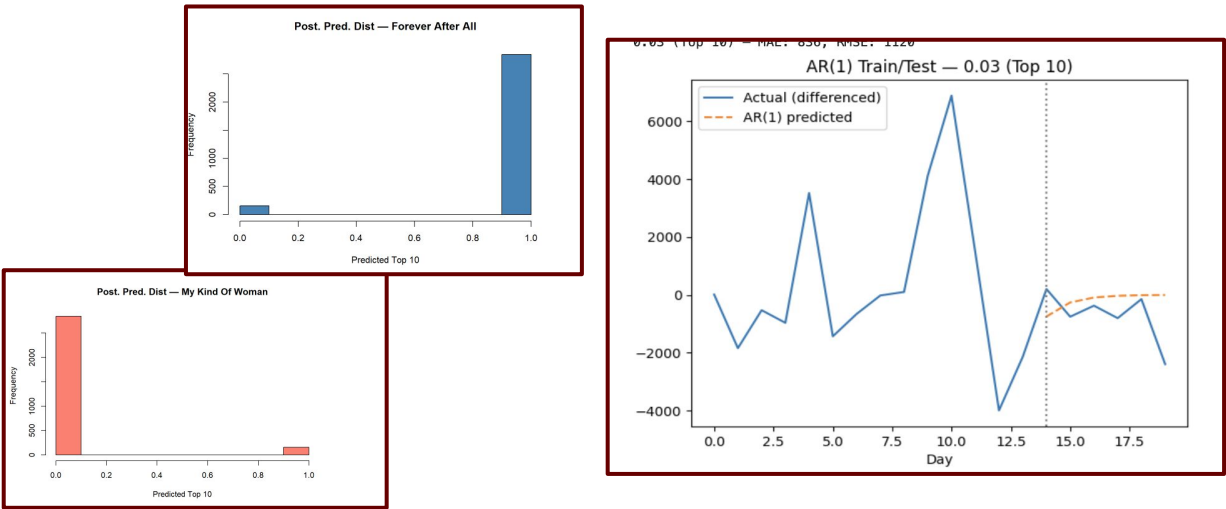
Bayesian Logistic Regression using **R**

- Predictors: trajectory features + audio attributes, standardized
- Priors: N(0,2) on coefficients, Student-t(4,0,5) on intercept
- MCMC: 4 chains, 2000 iterations, 500 warmup

RESULTS

AR/MA Findings

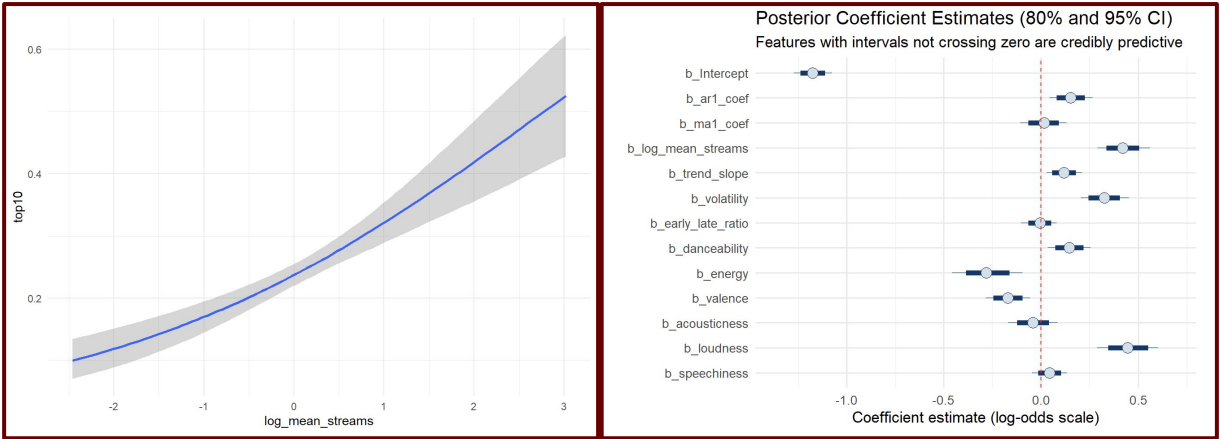
The AR/MA pipeline processed 3,253 songs with a fixed ARMA(1,1) order. Class distribution: 2,374 non-hits (73%) vs 879 Top 10 hits (27%).



Bayesian Findings

The Bayesian model achieved a test accuracy of 72.7% on the held-out set (n=651). The confusion matrix reveals 438 true negatives and 34 true positives with only 14 false positives, indicating high precision.

## Regression Coefficients:							
##	Estimate	Est.Error	1-95% CI	u-95% CI	Rhat	Bulk_ESS	Tail_ESS
## Intercept	-1.17	0.05	-1.27	-1.08	1.00	2864	2588
## ar1_coef	0.15	0.06	0.04	0.26	1.00	2865	2701
## ma1_coef	0.01	0.06	-0.11	0.13	1.00	2866	2626
## log_mean_streams	0.42	0.07	0.29	0.56	1.00	2750	2691
## trend_slope	0.12	0.05	0.03	0.21	1.00	2485	2549
## volatility	0.33	0.06	0.21	0.45	1.00	2923	2637
## early_late_ratio	-0.01	0.05	-0.10	0.08	1.00	2838	2891
## danceability	0.15	0.06	0.03	0.26	1.00	2405	2574
## energy	-0.28	0.09	-0.46	-0.09	1.00	2995	2990
## valence	-0.17	0.06	-0.28	-0.06	1.00	2887	2518
## acousticness	-0.04	0.06	-0.17	0.09	1.00	3039	2489
## loudness	0.45	0.08	0.29	0.60	1.00	2668	2756
## speechiness	0.04	0.05	-0.05	0.13	1.00	2872	2553



CONCLUSIONS

Trajectory-derived features add meaningful predictive value beyond audio attributes alone.

Key findings:

- Loudness is the strongest audio predictor so louder songs more likely to chart
- Log_mean_streams is the strongest predictor overall, as higher early stream volume strongly increases P(Top10) more than any audio feature
- Energy and valence are negatively associated which is surprising but consistent with recent trends toward mellower hits
- Ar1_coef and volatility are credibly positive, confirming trajectory in the first 3 weeks a valid signal

The main limitations are class imbalance (73% non-hits) which biases predictions toward the negative class, a low dataset join rate (~6%) limiting sample size.

NEXT STEPS

- Handle class imbalance via weighted likelihood or oversampling to improve recall on Top 10 hits
- Extend to multi-class prediction like Top 10 / Top 50 / Top 100 / Below
- Add artist-level random effects to account for popularity variation across artists

REFERENCES

[1] Cabansag, I. J., and Ntegeka, P. (2025). "Prediction of Spotify Chart Success Using Audio and Streaming Features." arXiv.
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